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11 **(a)** **(i)** State the name of **one** planet that has an orbit further away from the Sun than Venus.

..... [1]

(ii) State the name of **one** planet that has an orbit closer to the Sun than Venus.

..... [1]

(b) Venus has an average radius of orbit of 1.1×10^{11} m and an orbital period of 220 Earth days. Calculate the average orbital speed of Venus. Give your answer in m/s.

average orbital speed = ... m/s [3]

(c) State the relationship between the orbital speeds of the planets and their distances from the Sun.

.....

..... [1]

(d) Comets are balls of ice and dust. Some comets orbit the Sun.

State how the speed of a comet changes as it orbits the Sun.
Explain your answer using ideas about the conservation of energy.

You may include a labelled diagram in your answer.

.....

.....

..... [3]

[Total: 9]